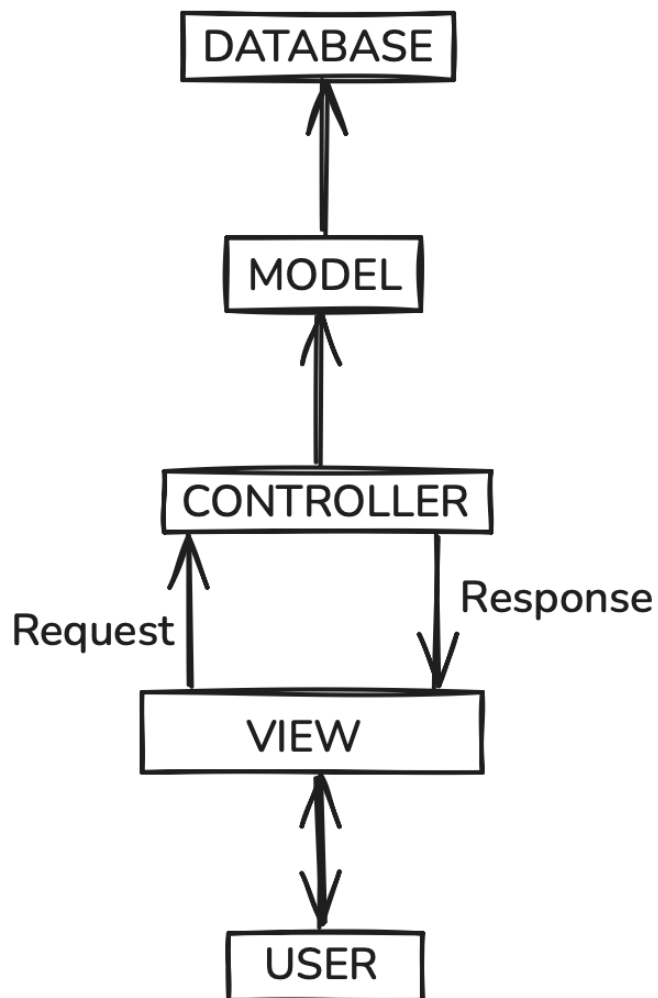


MVC Pattern

Description:- The House Rent application follows the **Model-View-Controller** (MVC) architectural pattern, which separates the application into three interconnected components: **Model**, **View**, and **Controller**. This structure improves code organization, maintainability, scalability, and development efficiency.

MVC Diagram



Model Layer (Data Layer)

The **Model Layer** is responsible for handling all application data and database operations. It defines the data structure and manages interactions with MongoDB using Mongoose schemas.

Models Used:

- User Model
- Property Model
- Booking Model
- Booking History Model

Responsibilities:

- Store user information
 - Manage property listings
 - Handle booking records
 - Maintain booking history
 - Perform CRUD operations on data
-

Controller Layer

The **Controller Layer** acts as an intermediary between the View and Model layers. It receives requests from users, processes the business logic, interacts with the appropriate models, and sends responses back to the client.

Responsibilities:

- Process user registration and login
 - Handle property search requests
 - Manage property listings
 - Process booking requests
 - Update booking status
 - Manage booking history records
 - Validate incoming data
-

View Layer (Presentation Layer)

The **View Layer** represents the user interface of the application. It is developed using React.js and is responsible for displaying data to users and collecting user inputs.

Responsibilities:

- Display property listings
- Show property details
- Provide search and filter functionality
- Display booking forms
- Show booking history
- Present user dashboards
- Display status notifications